Performance Work Statement (PWS)
For Internal Communications System (ICS)
Engineering, Technical and Fabrication Services

1.0 Background

The Undersea Warfare (USW) Electromagnetic Systems Department (Code 34) of the Naval Undersea Warfare Center Division, Newport (NUWCDIVNPT) performs engineering, technical, acquisition, fabrication, and program management functions in support of Communication Systems and Electromagnetic Maneuver Warfare (EMW) programs. Program Manager Ships (PMS) 392 is the Participating Acquisition and Resource Manager (PARM) for all attack submarine platforms and has designated Code 34 as In-Service Engineering Agent (ISEA) for the Virginia class submarine Integrated Communications Systems (ICS). Under the direction of PMS-392, NUWCDIVNPT Code 34 provides system and subsystem development, procurement, integration, testing, and in-service support. Additionally, PMS-450, PMS-397, and the Fleet task Code 34 with engineering design and in-service support tasks for the Virginia and Columbia ICS systems.

1.1 Places of Performance

The work specified herein will be performed at NUWCDIVNPT, the Contractor's facility, and other locations, as identified in individual Technical Instructions (TIs). It is anticipated that the Contractor may travel to the following locations during performance:

- Norfolk, VA
- Groton, CT
- Charleston, SC
- Washington, DC
- Pearl Harbor, HI
- San Diego, CA
- Guam
- Yokosuka, Japan
- Kingsbay, GA
- Bremerton, WA
- Naples, Italy
- Shipboard and at sea testing (various locations)

1.2 Authorized Users

The USW Electromagnetic Systems Department, Code 34, is the only authorized user.

1.3 Sponsors

The Government anticipates the following sponsors will fund this contract: PEO Submarines (SUB) PMS-397, 450, 392, 396

1.4 Types of Funding

The following types of funding are anticipated to be obligated under this contract:

- Operation and Maintenance, Navy (OMN)
- Other Procurement, Navy (OPN)

- Shipbuilding and Conversion, Navy (SCN)
- Working Capital Funds (WCF)

2.0 Scope

This tasking is for engineering, technical and fabrication services for the Integrated Communications System (ICS) used onboard Virginia and Columbia Class Submarines. The ICS (hereinafter referred to as "System") is comprised of the ICS cabinet, racks, and components, cabling, distributed communications equipment, wireless communications equipment, software as well as interfaces with other associated systems and equipment.

Obsolete parts and parts experiencing atypical failures are hereinafter referred to as obsolete items.

The contractor shall investigate, research, troubleshoot, analyze, and resolve problems resulting from obsolete items; and design, fabricate, test, integrate, and install remanufactured, redesigned, or modernized replacements for obsolete items.

3.0 Applicable Documents

The Contractor shall perform the tasking required in Section 4.0 in accordance with the below Applicable Documents (ADs)

Number	Title
3.1	DoD Issuances
3.1.1	MIL-STD-1399C Interface Standard for Shipboard Systems, 02 February 1988
3.1.2	MIL-STD-31000B Technical Data Packages, 31 October 2018
3.1.3	MIL-STD-882E Standard Practice for System Safety, 11 May 2012
3.1.4	MIL-D-23140D Drawings, Installation Control, For Electronic Equipment, 30 April 1992
3.1.5	MIL-DTL-901E Requirements For Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, And Systems, 20 June 2017
3.1.6	MIL-STD-461G Requirements for Control of Electromagnetic Interference, Characteristics of Subsystem and Equipment, 11 December 2015
3.1.7	MIL-STD-167-1A Mechanical Vibrations of Shipboard Equipment (Type I-Environmental and Type II-Internally Excited), 02 November 2005
3.1.8	MIL-STD-740 Structure borne Vibratory Acceleration Measurements Acceptance Criteria of Shipboard Equipment, 30 December 1986
3.1.9	MIL-STD-810H Environmental Engineering Considerations and Laboratory Tests, 31 January 2019
3.1.10	MIL-STD-1310 Shipboard Bonding, Grounding, And Other Techniques For Electromagnetic Compatibility, Electromagnetic Pulse (Emp) Mitigation, and Safety, 12 August 2014
3.1.11	MIL-HDBK-454 General Guidelines for Electronic Equipment, 30 June 2020
3.1.12	MIL-STD-1683 Connectors And Jacketed Cable, Electric, Selection Standard For Shipboard Use, 31 March 1989
3.1.13	DoD 8570.01-M, Information Assurance Workforce Improvement Program, Change 4, 10 November 2015

3.1.14	DoDM 5200.01, Vol 1, DoD Information Security Program, Change 3, 17 January 2025
3.1.15	DoDM 5200.01, Vol 2, DoD Information Security Program, Change 4, 28 July 2020
3.1.16	DoDM 5200.01, Vol 3, DoD Information Security Program, Change 4, 17 January 2025
3.1.17	DoDM 5200.48 Controlled Unclassified Information, 06 March 2020
3.2	Navy Issuances
3.2.1	NAVSEA S9070-AA-MME-010/SSN/SSBN, Technical Requirements Manual For Temporary Submarine Alterations (ACN-5) Revision 3, 28 March 2018
3.2.2	NAVSEA 0902-LP-018-2010 General Overhaul Specifications for Deep Diving SSBN/SSN Submarines (DDGOS), Revision B, 1 August 2018
3.2.3	NAVSEA S9407-AB-HBK-010 Handbook of Shipboard Electromagnetic Shielding Practices (Rev 2 Chg. 1), 22 February 2020
3.2.4	MS No. 6310-081-015 SUBMEPP Submarine Maintenance Standard, MS No. 6310-081-015 Revision D; Submarine Preservation, 16 August 2016
3.2.5	Submarine Non-Propulsion Electronic System (NPES) Environmental Qualification Test (EQT) Requirements (Class Platform Unique And Common) Specification For Inboard Equipment, 23 January 2014
3.2.6	Life Cycle Configuration Management Implementation Manual (LCCMIM), February 2007
3.2.7	Participating Manager (PARM) Alteration Format Style Instruction, 50225-04987580, 24 May 2016
3.2.8	Submarine Communications Production Drawing Format Style Instruction, 50225-04987581, 24 May 2016
3.2.9	NUWCDIVNPTINST 4000.1B Receipt, Delivery, Inspection, and Shipment of Material, 11 September 2023
3.2.10	NAVSEA M-5510.1, NAVESA Security Program Manual, July 2022
3.2.11	NSWCHQ/NUWCHQINST 5510.1, Security Program Manual, 09 November 2020
3.2.12	NUWCDIVNPT 5500.4, NUWCDIVNPT Command Security Manual, 19 April 2023
3.3	Industry Standards
3.3.1	ASME Y14.100-2017 Engineering Drawing Practices
3.3.2	ASME Y14.24-2012 Types of Application of Engineering Drawings
3.3.3	ASME Y14.35M-2017 Engineering Drawings and Associated Documents
3.3.4	ASME Y14.5-2009 Dimensioning and Tolerancing
3.3.5	ASME Y14.38 Abbreviations and Acronyms for Use on Drawings and Related Documents
3.3.6	ASME Y14.34 Associated Lists
3.3.7	ISO-9001:2015; International Organization for Standardization (ISO) Quality Management Systems -- Requirements 9001:2015, 23 September 2015

4.0 Technical Requirements

The Contractor shall accomplish the following tasks in support of System problems and failures caused by obsolete items as detailed further at the Task Order level. As further defined at the task order level when severable services task orders are issued, Technical Instructions (TIs) will be issued to initiate specific tasking as particular needs and obsolete items are identified. For both severable (level of effort) and non-severable (job unit based) task orders, the Contractor shall perform the tasks in Section 4.0 in accordance with TIs and ADs (Section 3.0), and using Government Furnished Information (GFI) (Section 6.0).

4.1 Engineering, Systems Engineering and Process Engineering

4.1.1 Technical Data and Failure Analysis, and System Repair

Upon receipt of a TI, the Contractor shall complete investigations, research, ship checks, tests, and studies to identify and assess potential need for replacement parts for obsolete items. The contractor shall identify differences between the obsolete items and the recommended alternatives, modernization upgrades. The contractor shall identify and submit recommendations to the Government for review and approval. The contractor's recommendations shall identify the changes, updates, and revisions to documentation , supply and logistics, and training protocols resulting from the replacement parts.

Using the information provided as GFI, the Contractor shall determine if the replacement part(s) meet the current system's performance and operational requirements, and when inadequacies and obsolesce concerns are likely to start adversely affect the System. The Contractor shall systematically consider the Government's requirements identified in the TI and complete a gap or requirements analysis to synthesize and evaluate alternative concepts, identifying all assumptions, strengths, and shortcomings of the capabilities. The Contractor shall prepare and deliver an Existing Systems Gap Analysis Report to include recommendations for the modification, upgrade, or modernization of obsolete items in existing Systems. Upon Government approval of the recommended solution, the Contractor shall generate a design and system specification that considers how the new design will be integrated into the existing system. The gap analysis shall include an evaluation of the current concept of operations, potential end of life or obsolesce risks and suitable replacements, and identify any areas that can be improved with the software and hardware interface to the operator.

Deliverables: The Contractor shall prepare and deliver:

- Existing Systems Gap Analysis Report (CDRL A001);
- SSS (CDRL A002);
- Operational Concept Description (CDRL A003);
- IDD (CDRL A004);
- ICD (CDRL A005);
- Performance Specification Documents (CDRL A006);
- SSDD (CDRL A007); and,
- Developmental Design Drawings/Models and Associated Lists (CDRL A008).

4.2 Prototyping, Model-Making, Fabrication, and Pre-Production

Upon receipt of a TI, the Contractor shall develop, fabricate, build, and assemble reduced and full-scale models, mock-ups, prototypes, pre-production units, special support equipment, and research and development (R&D) test tools for replacement parts in accordance with Government approved drawings provided as GFI. The Contractor shall submit a Request for Waiver (RFW) or Request for Variance (RFV) in order to deviate or depart from any Government approved drawings. The contractor shall provide redlined engineering drawings and Waiver or Deviation System Safety Report with RFW and RFV submissions. The Contractor shall not complete any work affected by the proposed waiver or variance without first receiving Government approval of the RFW or RFV. The Contractor shall prepare and provide for Government approval Factory Acceptance Test Plans and Procedures to test that the models, mock-ups, prototypes, pre- production units, special support equipment, and R&D test tools meet the Government's performance specifications for the replacement parts and overall System provided as GFI. Upon Government approval of the Test Plans and Procedures, the Contractor shall test, operate, and evaluate the reduced and full-scale models, mock-ups, prototypes, special support equipment, research and development (R&D) test tools, and pre-production replacement parts. The Contractor shall prepare and deliver a Technical Data Package and Acceptance Test Report for every testing series. The contractor shall prepare and deliver Certificates of Compliance and Certification and Certification Data Reports or Acceptance Test Reports. The Contractor shall prepare and provide for Government review and approval, an Assessment of Manufacturing Risk and Readiness, prior to starting fabrication. Upon Government approval, the Contractor shall perform fabrication services. The Contractor shall notify the Government of errors and required changes to GFI immediately upon discovery of any issues. The Contractor shall include source material for the deliverables.

Deliverables:
Hardware: Full scale models, mock-ups, prototypes, and pre-production Systems.

Data: The Contractor shall prepare and deliver:
- Request for Waiver (CDRL A00B);
- Request for Variance (CDRL A00C);
- Waiver or Deviation System Safety Report (WDSSR) (CDRL A00D);
- Certificates of Compliance (CDRL A00E);
- Engineering Drawings (CDRL A00F);
- Technical Data Package (CDRL A00G);
- Certification/Data Report (CDRL A00H);
- Test Plans/Test Procedures (CDRL A00J);
- Acceptance Test Report (CDRL A00K); and,
- Assessment of Manufacturing Risk and Readiness (CDRL A00L).

4.3 System Design Documentation and Technical Data

4.3.1 Interoperability, Test and Evaluation and Trials Support Documentation

Upon receipt of a TI, the Contractor shall develop and update Factory Acceptance Test Plans and procedures and Test and Evaluation Program Plans that address the scope, methods, and steps for conducting interoperability testing, test and evaluation, trials and data collected for replacement parts. The Contractor shall recommend the factory acceptance test plan acceptable pass/fail criteria in accordance with performance parameters provided as GFI in an individual TI. Upon issuance of Government approved acceptable pass/fail criteria, the Contractor shall prepare and deliver the final factory acceptance test plans and procedures.

Deliverables: The Contractor shall prepare and deliver:
- Acceptance Test Plan (CDRL A00M);
- Test Plans/Test Procedures (CDRL A00J);
- Test and Evaluation Program Plan (TEPP) (CDRL A00N); and,
- Equipment Installation Instructions (CDRL A00P)

4.3.2 Drawing Updates

Upon receipt of a TI, the Contractor shall review and revise Government approved redline drawings and documentation identified in the TI to reflect replacement parts resulting from government approved changes.

Deliverables: The Contractor shall prepare and deliver:

- SSS (CDRL A002);
- Operational Concept Description (CDRL A003);
- IDD (CDRL A004);
- Engineering Drawings (CDRL A00F);
- Design Data and Calculations (CDRL A00Q);
- Technical Data Package (CDRL A00G);
- Engineering Change Proposals (ECP) (CDRL A00R);
- Engineering Change Proposal System Safety Report (ECPSSR) (CDRL A00S);
- Notice of Revision (NOR) (CDRL A00T);
- Technical Data Package Review Report (CDRL A00U);
- Equipment Installation Instructions (CDRL A00P); and,
- Certification/Data Report (CDRL A00H).

4.4 Software Engineering, Development, Programming, and Networks

4.4.1 Software Requirement Specifications

Upon receipt of a TI, and in accordance with NAVSEA Text C-227-H007, the Contractor shall revise and deliver the Software Development Plan (SDP). The Contractor shall follow the Government concurred-with SDP for all computer software to be developed or maintained under this effort. The Contractor shall update the SDP throughout the life of the contract. The Contractor shall develop, adapt, and modify Software Requirement Specifications (SRS), Software Design Descriptions (SDD), System Requirements Verification Matrix (SRVM), Database Design Documents (DBDD), Software Test Plans (STP), Software Test Descriptions (STD), Software Test Reports, Software Version Description (SVD) and Software Product Specifications (SPS) in support of software development of production, simulation, and prototype code to support the introduction and use of replacement parts.

Deliverables: The Contractor shall prepare and deliver:

- SPD (CDRL A00V);
- SRS (CDRL A00W);
- SDD (CDRL A00X);
- DBDD (CDRL A00Y);
- STP (CDRL A00Z);
- STD (CDRL A010);
- SPS (CDRL A011); and,
- SVD (CDRL A012).

4.4.2 Analysis, Development and Selection of Hardware and Software

Upon receipt of a TI, the Contractor shall review SRS, SDD, SRVM, DBDD, STPs, STDs, Software Test Reports, SPS, and software source code for the system and components to validate that the replacement parts meet the system specifications provided as GFI with applicable TIs. The Contractor shall design and develop software code and executable software modules for the use of replacement parts in the System. The Contractor shall develop test code and simulators software and conduct software testing.

Deliverables: The Contractor shall prepare and deliver:

- SSS (CDRL A00W);
- Operational Concept Description (CDRL A003);
- IDD (CDRL A004);

- Computer Software Product (CDRL A013);
- Computer Software Product End Items (CDRL A014);
- Computer Program End Item Documentation (CDRL A015);
- SRS (CDRL A00W);
- SDD (CDRL A00X);
- DBDD (CDRL A00Y);
- STP (CDRL A00Z);
- STD (CDRL A010);
- STR (CDRL A016);
- SPS (CDRL A011);
- SVD (CDRL A012); and,
- Computer Operation Manual (COM) (CDRL A017).

4.4.3 Cyber Security (CS) and Anti-Tamper (AT)

Upon receipt of a TI, and related solely to ICS products, the Contractor shall provide qualified CSWF personnel for the development, operation, management, and compliance of Cyber Security policies and implementation of security measures and procedures for Government Information Systems. The Contractor shall review and validate, as a Navy Qualified Validator, Risk Management Framework (RMF) packages provided as GFI and deliver validated RMF packages to the Government for final review and approval. The Contractor shall maintain and update Information Assurance (IA) security planning and accreditation documentation. The Contractor shall meet and maintain the certification and educational requirements required by the Cyber Security Work Force and associated with Network Services (Specialty Code 44, Expert Level), System Administration (Specialty Code 45, Expert Level), and Systems Security Analysis (Specialty Code 46, Expert level). The Contractor shall conduct test events as part of the Assessment and Authorization (A&A) process.

The Contractor shall conduct scans of the Government System network aligning with SWFTS accreditation package scheduling, monthly for non-SWFTS systems, or as defined in the TI. The Contractor shall use Assured Compliance Assessment Solution (ACAS) to conduct the scans and shall provide the results to the Government. The Contractor shall maintain the ACAS scanning tools with the latest plugins, benchmarks, and scanning engines prior to the execution of each vulnerability scan. Utilizing the scan results, the Contractor shall document and maintain the following in the ACAS Report: Information Assurance Vulnerability Alerts (IAVA) compliance, Information Assurance Vulnerability Bulletins (IAVB), and Security Technical Information Guides (STIG) within the Systems.

The Contractor shall conduct studies and analyses of new IA and IT trends, policies and regulations and investigate the feasibility of applying new solutions into the infrastructure and architecture. The Contractor shall prepare and deliver Cyber Security/Anti- Tamper Documentation to provide recommendations for improvement in IA hardware, software and peripherals to the Government for consideration for future upgrades. The Contractor shall implement cybersecurity plans in accordance with the Government-approved program protection plan and attend and participate in Government cyber test and evaluation events.

Deliverables: The Contractor shall prepare and deliver:

- ACAS Report (ACAS) (CDRL A018);
- DOD Risk Management Framework (RMF) Package Deliverables (CDRL A019); and,
- CSAT (CDRL A001).

4.4.4 Cyberspace Qualifications

Per DFARS 252.239-7001 'Information Assurance Contracting Training and Certification,' all personnel performing cyber functions must be trained and qualified. In addition, personnel shall maintain the appropriate security clearance to perform the tasks associated with their assigned positions.

Note: DFARS uses Information Assurance (IA) which is the old title and acronym; it has been replaced by Cyberspace/Cyber Workforce (CWF).

The contractor is required to earn and maintain appropriate baseline certification as identified in DOD 8570.01-M for the position and tasking being performed.

If privileged access is required, the contractor shall complete a privileged access agreement, SECNAV 5239/1, and submit it to the COR. Contractors have up to 6 months to obtain applicable operating system and computing environment training or the account(s) will be suspended. The contractor shall provide a copy of the certificate(s) of completion to the COR.

4.4.5 Cyberspace Reporting

The contractor shall provide a list of all personnel assigned to the task order performing cyber functions as a part of a monthly CWF Report. The report shall include employee name, task order number, the applicable CWF category and level, required certifications and fulfillment status. Note that Contractor CWF personnel must receive designation approval in writing by the Commanding Officer, or the C-ISSM via "By Direction" authority, prior to performing CWF functions.

The contractor shall submit new hire information for tasking requiring cyber functions to the COR at least seven (7) days prior to the new hire employee beginning performance of any cyber functions. New hire information shall include name, task order number, list of applicable cyber functions category, level, required certifications and fulfillment status, as well as a copy of the baseline certification documentation. Per DFARS 252.239-7001(c), "Contractor personnel who do not have proper and current certifications shall be denied access to DoD information systems for the purpose of performing information assurance functions, and therefore may not be allowed to charge to the task order.

Deliverable: The Contractor shall deliver a monthly CWF Report in accordance with CDRL A00X.

4.5 Reliability and Maintainability

Upon receipt of a TI, the Contractor shall conduct engineering, scientific, and analytical studies of the System to analyze, evaluate, and design Reliability and Maintainability (R&M) requirements for replacement parts. The Contractor shall integrate the System R&M requirements with the System design, development and life- cycle sustainment. Upon Government approval of the R&M design requirements, the Contractor shall provide an R&M Test Plan for Government approval. Upon approval of the R&M Test Plan, the Contractor shall detect and correct System design deficiencies, failed parts, and workmanship defects that affect functionality. After completing the repairs, the contractor shall conduct reliability testing and provide a Reliability Test Report. In the R&M analysis, the Contractor shall address the mission reliability and logistics reliability probability that the System will function properly under the stated operating conditions for the System identified in a TI.

Deliverables: The Contractor shall prepare and deliver:

- Reliability & Maintainability Analysis (CDRL A001);
- R&M Program Plan (CDRL A01A);
- Reliability-Centered Maintenance Analysis Data (CDRL A01B);
- Failure Modes Effects and Criticality Analysis (CDRL A01C);
- Maintenance Task Analysis (CDRL A01D);

- Reliability Qualified Items List (CDRL A01E);
- Reliability and Maintainability Test Plan (CDRL A01F); and,
- Reliability Test Report (CDRL A01G).

4.6 Interoperability, Test and Evaluation, and Trials

4.6.1 Interoperability Testing Studies

Upon receipt of a TI, using Government approved test plans and procedures identified in individual TIs, the Contractor shall conduct studies and testing to validate that Systems with replacement parts meet joint interoperability requirements at all levels of their life cycle. The Contractor shall prepare and deliver Interoperability Reports.

Deliverables: The Contractor shall prepare and deliver:

- Interoperability Report (CDRL A01S).

4.6.2 Testing

Upon receipt of a TI, the Contractor shall develop and update test plans, procedures, and reports for Systems with replacement parts. The contractor shall review test plans and procedures prior to conducting the test. Following Government approval of test plans and procedures, the Contractor shall conduct land-based, dockside, environmental, or at-sea testing. The Contractor shall record the results of the testing and provide recommendations for design changes or improvements to the Government. The Contractor shall provide these observations, records, and recommendations in technical reports. The Contractor shall deliver a completed Test Plan and Test Procedure. The Contractor shall deliver Trip/Travel Reports upon completion of travel. When conducting testing the contractor shall recommend redlines to test procedures.

Deliverables: The Contractor shall prepare and deliver:

- Testing Data Report (CDRL A001);
- Test/Inspection Report (CDRL A01S);
- Test Plan (CDRL A01T); Test Procedure (CDRL A01U);
- Acceptance Test Plans (CDRL A00M);
- Test Plans/Test Procedures (CDRL A00J);
- Acceptance Test Report (CDRL A00K);
- Performance Analysis Report (CDRL A009);
- Test and Evaluation Program Plan (TEPP) (CDRL A00N); and,
- Trip/Travel Reports (CDRL A01V).

4.7 Training

4.7.1 Training Recommendations

Upon receipt of a TI, the Contractor shall conduct studies and provide recommendations for the development of training curricula for the operation and maintenance of the System with replacement parts. As part of the training recommendation, the Contractor shall include applied exercises resulting in the attainment and retention of knowledge, skills, and abilities regarding the platforms, Systems, and warfighting capabilities.

Deliverables: The Contractor shall prepare and deliver:

- Training Program Development and Management Plans (CDRL A01W).

4.7.2 Training Curricula Development

Upon receipt of a TI, the Contractor shall develop and update training plans, course curricula, and training documentation for Systems with replacement parts. The Contractor shall develop instructional material for initial installations, engineering changes, operations, and maintenance of Systems. The Contractor shall include: general, physical, and functional descriptions; system operations; and system operational level maintenance in the training material.

Deliverables: The Contractor shall prepare and deliver:

- Training Materials (CDRL A01X);
- Training Conduct Support Document (CDRL A01Y);
- Lesson Strategy Report (CDRL A01Z);
- Student Guide (CDRL A020);
- Training Equipment Summary (CDRL A021); and,
- Training Evaluation Document (CDRL A022).

4.7.3 Conduct Training

Upon receipt of a TI, the Contractor shall provide all required Government-approved training materials and instructors for training courses for Systems with replacement parts. The Contractor shall conduct classroom courses and demonstration sessions in the areas of proper and safe operation, maintenance, and employment of Systems with replacement parts. The Contractor shall conduct training shipboard and at land-based installations. Upon completion of trainings, the Contractor shall provide completed Training Evaluation Documents and Trip/Travel Reports.

Deliverables: The Contractor shall prepare and deliver:

- Training Evaluation Document (CDRL A022); and,
- Trip/Travel Reports (CDRL A01V).

4.8 In-Service Engineering Fleet Introduction, Installation and Checkout

4.8.1 Technical Data Analysis

Upon receipt of a TI, and in accordance with ADs identified in the TI, the Contractor shall collect technical data identified in the TI and analyze results, identify issues, and provide recommendations to resolve the issues with obsolete items. The Contractor shall prepare and deliver technical reports for each tested System, subsystem, or component.

Deliverables: The Contractor shall prepare and deliver:

- ISEA Technical Data Analysis Report (CDRL A001); and,
- Trip/Travel Reports (CDRL A01V).

4.8.2 System Troubleshooting and Repair

Upon receipt of a TI, the Contractor shall troubleshoot and identify and isolate defects of the System. The Contractor shall provide an estimate of required repairs to the Government. Upon Government approval of the repair estimate, the Contractor shall conduct repairs and testing of System to verify that repairs are completed in accordance with performance specifications and testing documentation identified in the TI. The Contractor shall prepare and deliver technical reports summarizing the completed repairs, testing results and findings, corrective actions taken and any outstanding problems or issues.

Deliverables: The Contractor shall prepare and deliver:

- System Repair Estimate and Report (CDRL A001);

- Certification/Data Report (CDRL A00H); and,
- Trip/Travel Reports (CDRL A01V).

4.8.3 Technical Investigation

Upon receipt of a TI, the Contractor shall conduct on-site or remote technical investigations on developmental and operational Systems in response to problems. The Contractor shall identify and document the possible causes of the reported problem(s) and recommend potential solutions to the Government.

Deliverables: The Contractor shall prepare and deliver:

- Technical Investigation Report (CDRL A001); and,
- Trip/Travel Reports (CDRL A01V).

4.8.4 Failure Analysis and Reporting

Upon receipt of a TI, the Contractor shall analyze System failures identified in the TI and provide failure analysis reports identifying trends, root cause failure modes, mechanisms, and parts in response to problems identified. The Contractor shall update the Failure Reporting Analysis and Corrective Action System (FRACAS) to track failure analysis reports.

Deliverables: The Contractor shall prepare and deliver:

- Failure Summary and Analysis Report (CDRL A023);
- Failure Analysis and Corrective Action Report (FACAR) (CDRL A024); and,
- Trip/Travel Reports (CDRL A01V).

4.8.5 Installations

Upon receipt of a TI, the Contractor shall prepare and deliver an Alteration Installation Team Quality Assurance Workbook and maintain the Workbook on-site during installations. The Contractor shall make the Workbook available to the Government upon request during the installation. The Contractor shall provide daily status updates to the On-Site Installation Coordinator (OSIC). The Contractor shall install TEMPALTs, OPALTs, SHIPALTS, ECIs, hardware, and software in laboratories or on vessels, as identified in the TI. The Contractor shall perform pre-installation testing, power up, checkout, and post-installation testing of installed Systems. The Contractor shall remove TEMPALTS. The Contractor shall generate and deliver DFS source material for all installation non-compliances. The Contractor shall develop and deliver completion messages immediately at the conclusion of the installation. The Contractor shall prepare and deliver Trip Reports.

Deliverables: The Contractor shall prepare and deliver:

- Alteration Installation Team Quality Assurance Workbook (CDRL A025);
- Test/Inspection Report (CDRL A01S);
- Installation Completion Message (CDRL A001); and,
- Trip/Travel Reports (CDRL A01V).

5.0 Progress Reports

The Contractor shall prepare a Contractor's Status Report that indicates the progress of work, status of the program(s), and existing or potential problem areas for all assigned tasks. The Contractor shall submit the Contract Status Report for the same timeframe as each invoice submitted in the Wide Area Workflow (WAWF) Module of the Procurement Integrated Enterprise Environment (PIEE).

The Contractor shall require each Contractor employee performing work at any NUWC Division Newport site, who have a Common Access Card (CAC) and an NMCI Account complete all required trainings. The required training and required training schedule is available at the following website: https://wiki.navsea.navy.mil/pages/viewpage.action?spaceKey=NCOW&title=NUWCDIVNPT+Workforce+Development+Program. See the pdf for Mandatory Training Instruction – Contractors Only.

The Contractor shall prepare and submit Training Completion Status Reports affirming all required training has been completed. The Training Completion Status Reports shall include the contract/order number, contactor employee name, specific training name, training course ID #, and completion date for all contractor employees asserting training completion. See the pdf for Mandatory Training Instruction – Contractors Only. Contact the COR identified in Section G if additional access instructions are needed. If specific training questions arise, contact the POC listed for the training in the pdf.

Deliverables: The Contractor shall prepare and deliver:

- Contract Status Report (CDRL A02A);
- Training Status Report (CDRL A02C).

6.0 Government Furnished Information

The following Government Furnished Information (GFI) will be made available under this contract:

NUMBER	TITLE	TASK NUMBER
6.1	MIL-STD-461G, Requirements for Control of Electromagnetic Interference, Characteristics of Subsystem and Equipment	4.1
6.2	MIL-STD-1399C, Interface Standard for Shipboard Systems	4.1
6.3	MIL-STD-961E, Defense and Program-Unique Specifications Format and Content	4.1
6.4	MIL-STD-31000B, Department of Defense Standard Practice, Technical Data Packages	4.1.1 - 4.1.3, 4.4.2, 4.4.3
6.5	NAVSEA INSTRUCTION 4720.14F, Temporary Alterations to Fleet Submarines	4.4.3
6.6	NAVSEA S9070-AA-MME-010/SSN/SSBN Technical Requirements Manual For Temporary Submarine Alterations	4.4.3
6.7	DoD Directive 8000.01, Management of DoD Information Enterprise	4.5
6.8	DoD Directive 8140.01 Cyberspace Workforce Management of 11 August 2015	4.5
6.9	DoD Instruction (DoDI) 8500.01 Cybersecurity	4.5
6.10	DoD Directive 8500.1, Information Assurance	4.5
6.11	DoDI 8500.2, Information Assurance Implementation	4.5

Additional GFI may be identified in individual TIs.

7.0 Government Furnished Property

The Government may provide Government Furnished Property (GFP) as listed in Attachment 18. The contractor shall adhere to DFARS Clause 252.245–7005 for the management and reporting of Government Property and shall use the GFP Module of the Procurement Integrated Enterprise Environment (PIEE) as specified in the clause. In addition, the Contractor shall prepare and submit a Government Property (GP) Inventory Report to capture GFP status IAW CDRL A02B.

Deliverable: Government Property (GP) Inventory and Forecasting Report (CDRL A02B)

8.0 Quality Surveillance and Performance Standards

The Government will conduct quality surveillance via various methods including formal and informal meetings, review of technical reports, review of monthly progress reports, and review of deliverables. Contractor performance will be evaluated in the areas of technical quality, cost control, schedule/timeliness, management, and regulatory compliance as follows:

- Technical quality: The Government will evaluate technical quality using the criteria defined in the Performance Requirements Summary (PRS) Table, Attachment 2 to the Task Order.
- Cost Control: The Government will evaluate the Contractor's effectiveness in forecasting, managing, and controlling actual costs in comparison to negotiated costs;
- Schedule/Timeliness: The Government will evaluate the Contractor's ability to meet negotiated milestones and delivery schedules. This includes the Contractor's ability to meet CDRL schedules with minimal variance.
- Management: The Government will evaluate the Contractor's ability to integrate and coordinate all activities needed to execute the Task Order.
- Regulatory Compliance: The Government will evaluate the Contractor's compliance with the terms and conditions of the Task Order relating to applicable regulations and codes.

9.0 Exception From Information and Communication Technology Accessibility Requirements

The Government has determined that this procurement is an exception to the Information and Communication Technology (ICT) Accessibility Standards (36 C.F.R. Part 1194, Appendix A). Notwithstanding that an exception exists, the Contractor may furnish items or services provided under this contract that comply with the ICT Accessibility Standards (36 C.F.R. Part 1194, Appendix A).

- Exception #1 national security for (name the national security system/program)

10.0 Security Compliance

10.1 Program Protection (Classified)

SECURITY: All Contractor personnel shall adhere to the Security provisions of 32 CFR Part 117 – National Industrial Security Program Operating Manual (NISPOM) and the security requirements of the attached DD254. While performing work at a Government Facility, Contractor personnel shall comply with the security regulations of the host facility. Applicable FAR, DFARS, NMCARS clauses, and NAVSEA text shall be adhered to in the performance of this contract. Security incidents shall be promptly reported through the companies Facility Security Officer (FSO), to the Contracting Officer's Representative (COR), Technical Point of Contact (TPOC), and the Cognizant Security Office to NUWCDIVNPT Security.

1. **<u>Communications Security (COMSEC)</u>**
   COMSEC access is required for the contractors to perform tasks 4.10 and 4.12.
2. **<u>Sensitive Compartmented Information (SCI)</u>**
   SCI is required for the contractor to perform tasks 4.1, 4.5, and 4.12.

3. **Foreign Government Information (FGI)**
   FGI is required for the contractor to perform all Section 4 tasks.
4. **Alternative Compensatory Control Measures (ACCM) Information**
   ACCM is required for the contractor to perform tasks 4.10 and 4.12.
5. **Joint Worldwide Intelligence Communications System (JWICS)**
   JWICS is required for the contractor to perform tasks 4.1, 4.5, and 4.12.

Controlled Unclassified Information (CUI) including Legacy FOUO and Covered Defense Information (meeting the definition of 48 CFR 252.204–7012(a)) generated and/or provided under this contract shall be marked and safeguarded as specified in DoD Instruction 5200.48, CUI available at: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520048p.PDF. Any product containing Covered Defense Information shall be assigned a distribution statement (distribution statements B through F) in accordance with DoDI 5230.24 (Distribution Statements on Technical Documents); and DoDI 5230.24, Enclosure 3 Procedures, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/523024p.pdf

INFORMATION SECURITY: If the work is performed at the Contractor's facility, the Contractor shall implement and maintain security procedures and controls to prevent unauthorized disclosure of classified information and controlled unclassified information (CUI) and to control distribution of CUI in accordance with National Industrial Security Program Operating Manual (NISPOM) codifying 32 CFR Part 117, NISPOM Rule, and SECNAV M-5510.36B. If the work is performed at the Government's facility, the Contractor shall comply with facility policy.

CUI INCIDENT REPORTING AND RESPONSE: The Contractor shall promptly report any unauthorized, inadvertent, or illegal release or disclosure of CUI to the Contracting Officer's Representative / Technical Point of Contact (TPOC), Procuring Contracting Officer, and the Security Office. Contractor personnel shall coordinate this effort through the relevant industry site FSO.

PUBLIC RELEASE: Any controlled unclassified information pertaining to this contract shall not be released for public dissemination, including posting to any social media sites such as Facebook or Twitter, unless it has been approved for public release by appropriate U.S. Government authority. Proposed public releases shall be submitted for approval prior to release through the appropriate U.S. Government Office.

10.2 Operations Security (OPSEC)

OPSEC is a process that identifies critical information to determine if friendly actions can be observed by adversary intelligence systems, determines if information obtained by adversaries could be interpreted to be useful to them, and then executes selected measures that eliminate or reduce adversary exploitation of friendly critical information.

The Contractor shall develop and implement, and update and maintain an OPSEC program to protect controlled unclassified and classified activities, information, equipment, and material used or developed by the Contractor and any subcontractor during performance of the contract. The Contractor shall be responsible for the subcontractor implementation of the OPSEC requirements. The Contractor developed OPSEC program may include Information Assurance and Communications Security (COMSEC). The OPSEC program shall be in accordance with National Security Presidential Memorandum (NSPM) 28, and at a minimum shall include:

1) Assignment of responsibility for OPSEC direction and implementation.
2) Issuance of procedures and planning guidance for the use of OPSEC techniques to identify vulnerabilities and apply applicable countermeasures.
3) Establishment of OPSEC education and awareness training. To include initial and annual OPSEC training.
4) Provisions for management, annual review, and evaluation of OPSEC programs.
5) Flow down of OPSEC requirements to subcontractors when applicable.

While performing aboard Government sites, the contractor shall: comply with all OPSEC instructions and policies; include OPSEC as part of its ongoing security awareness program and take all required Agency training; Be responsive to the Supporting OPSEC Manager on a non-interference basis; and Protect sensitive unclassified information and activities, which could compromise classified information or operations, or degrade the planning and execution of operations performed by the Requiring Organization and contractor in support of the mission.

10.3 Electronic Spillage

Electronic Spillage (ES) is defined as a situation where information of higher classification than a system is authorized to process is introduced into that system, intentionally or otherwise. If a Contractor is determined to be responsible for an ES, all direct and indirect costs incurred by the Government for ES remediation will be charged to the Contractor.